19.3.1 Lamarckism

Jean Baptiste Lamarck (1744-1829) was a French naturalist, well known for his theory of evolution. Lamarck's theory of evolution was published in '**Philosophic Zoologique'** in the year 1809. It is popularly known as '**Theory of inheritance of Acquired Characters**" or "**Use and Disuse theory**" or **Lamarckism**.

Principles of Lamarckism

i. **Internal vital force**
Living organisms or their component parts tend to increase in size continuously. This increase in size is due to the inherent ability of the organisms.

ii. **Environment and new needs**
A change in the environment brings about changes in the need of the organisms. In response to the changing environment, the organisms develop certain adaptive characters. The adaptations of the organisms may be in the form of development of new parts of the body.

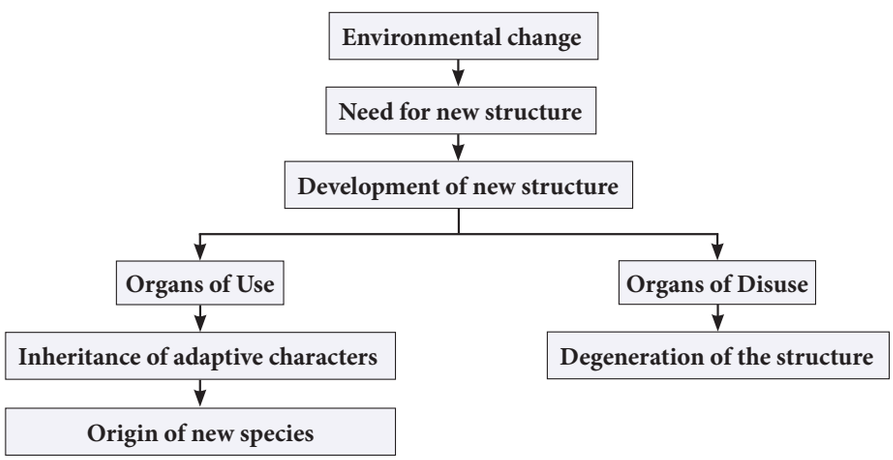
iii. **Use and disuse theory**
Lamarck's **use and disuse theory** states that if an organ is used constantly, the organ develops well and gets strengthened. When an organ is not used for a long time, it gradually degenerates.

The ancestors of giraffe were provided with short neck and short forelimbs. Due to shortage of grass, they were forced to feed on leaves from trees. The continuous stretching of their neck and forelimbs resulted in the development of long neck and long forelimbs which is an example for constant use of an organ. The degenerated wing of Kiwi is an example for organ of disuse.

iv. **Theory of Inheritance of acquired characters**
When there is a change in the environment, the animals respond to the change. They develop adaptive structures. The characters developed by the animals during their life time, in response to the environmental changes are called **acquired characters**. According to Lamarck, the acquired characters are transmitted to the offspring by the process of inheritance.

19.3.2 Darwinism or Theory of Natural Selection

Charles Darwin (1809-1882) was one of the great naturalist and philosopher of 18th century. He was born in England in 1809. While studying in college through his friendship with Professor J.S.Henslow he was fascinated towards nature. At that time the British Admiralty planned a **voyage of exploration** for 5 years on a ship named **H.M.S. Beagle** around **South America**. Dr Henslow was asked to nominate a young naturalist for the voyage. Darwin was given the opportunity. During his five years (1831–1835) voyage he visited many parts of



Flow chart showing the postulates of Lamarckism

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